Mahdieh Rojhannezhad, Ph.D.

GENETICS | CANCER & FUNCTIONAL GENOMICS | MOLECULAR DIAGNOSTICS | COMPUTATIONAL GENOMICS | LABORATORY QUALITY

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Professional Summary

Ph.D. geneticist with professional experience in cancer functional genomics, CRISPR/Cas9, molecular and cytogenetic diagnostics, laboratory supervision, and scientific communication. Doctoral research investigated a HER2-associated regulatory enhancer and lncRNA GAS5 in breast cancer. Complementary, project-based computational work includes NGS/WES workflows, genomic variant interpretation, viral-genomics analysis, RNA-seq, and molecular-dynamics workflows. Laboratory-quality knowledge combines prior diagnostic-laboratory responsibilities with self-directed study; no formal certification or independent regulatory practice is claimed.

Core Expertise

Cancer biology and regulatory genomics • CRISPR/Cas9 genome editing • Molecular diagnostics and cytogenetics • PCR, RT-PCR and qPCR • DNA/RNA extraction • Cell culture • Genomic variant interpretation • Laboratory workflow coordination • Scientific writing and teaching • QC documentation and data integrity

Professional Experience

Senior Specialist — Genetic Diagnostic Laboratories

Multiple laboratories, Iran | 2014–2023

• Performed molecular and cytogenetic analyses, including DNA/RNA extraction, PCR-based testing, and cell culture.

• Contributed to protocol optimization and research-oriented biotechnology initiatives.

Supervisor — Molecular & Cytogenetic Departments

Watson Medical Genetics Laboratory, Tehran, Iran | 2015–2019

• Led a team of 5–6 laboratory professionals and coordinated diagnostic laboratory operations and technical workflows.

• Oversaw PCR, RT-PCR, qPCR, karyotyping, and DNA/RNA extraction, with responsibility for quality and reporting accuracy.

Lecturer — Genetic Counseling

University of Rehabilitation Sciences & Social Welfare, Mashhad, Iran | 2014–2016

• Taught genetics and genetic counseling; mentored student research projects and scientific presentations.

Education

Ph.D., Genetics — Tarbiat Modares University, Tehran, Iran | 2016–2023 | Final grade: 19.5/20
Dissertation: Functional and bioinformatic investigation of HER2-associated enhancer GH17J039694 using CRISPR/Cas9 in breast-cancer cell lines.

M.Sc., Genetics — University of Tabriz, Iran | 2009–2011 | Final grade: 19.5/20
Thesis: Survivin –31 G/C promoter polymorphism and breast-cancer risk in East Azerbaijan, Iran.

B.Sc., Biology — University of Tabriz, Iran | 2002–2007

Selected Research & Computational Projects

HER2 enhancer functional genomics
CRISPR/Cas9-based investigation of a HER2-associated enhancer and lncRNA GAS5 in SKBR3, MCF7, and HEK293 cell lines, integrating molecular experiments with bioinformatic interpretation. Findings published in peer-reviewed journals.

Whole-exome sequencing workflow — project-based
Nextflow DSL2 workflow using fastp, BWA-MEM, SAMtools, GATK, and VEP with a GRCh38 reference. GitHub repository.

Viral genomics workflow — project-based
Reproducible Nextflow workflow for public Illumina Ebola virus data (SRR1553428), with QC, alignment, variant calling, and consensus generation; repository reports 100% genome coverage and 561 SNPs. GitHub repository.

TP53–PRIMA-1 molecular-dynamics project — work in progress
Docking-to-MD workflow using AutoDock Vina and GROMACS. Project snapshot: Zenodo DOI.

Additional projects: RNA-seq differential expression • WES variant interpretation case study • PRS literature review.

Laboratory Quality & Regulatory Knowledge

Self-directed study complementing prior diagnostic-laboratory quality responsibilities; familiarity rather than formal certification or independent regulatory practice.

• Frameworks: ISO/IEC 17025, ISO 15189, GLP, GMP; introductory knowledge of 21 CFR Part 11 and ALCOA+.

• QC and documentation: Westgard rules, Levey–Jennings charts, IQC/EQA, SOP principles, and document control.

• Improvement and validation: CAPA, root-cause analysis, internal-audit principles, calibration, IQ/OQ/PQ, method validation, and verification.

Technical Skills

Wet laboratory: PCR, RT-PCR, qPCR, CRISPR/Cas9, gRNA/primer design, molecular cloning, transfection, DNA/RNA extraction, mammalian cell culture, karyotyping, and gel electrophoresis.

Computational (project-based): WES/NGS workflows, Nextflow DSL2, BWA-MEM, GATK, BCFtools, VEP, GROMACS, AutoDock Vina, Linux/WSL2, Docker, Git/GitHub, Python-assisted analysis, and R data exploration.

Scientific tools: PyMOL, Cytoscape, BioRender, Adobe Illustrator, and Microsoft Office.

Selected Publications

• Rojhannezhad M, Soltani BM, Vasei M, Ghorbanmehr N, Mowla SJ. Functional analysis of a putative HER2-associated expressed enhancer, Her2-Enhancer1, in breast cancer cells. *Scientific Reports*. 2023;13:19516. DOI: 10.1038/s41598-023-46460-x.

• Rojhannezhad M, Mowla SJ. Gene Editing for Unraveling the Regulatory Role of a HER2-Associated Enhancer with lncRNA GAS5 and Related Genes in Breast Cancer Cells. *Journal of Advanced Immunopharmacology*. 2024;4(1):e142988.

• Rojhannezhad M, Hoseinpourfeizi MA, Pouladi N. The association between Survivin –31 G/C promoter polymorphism and breast cancer risk. *Zahedan Journal of Research in Medical Sciences*. 2015;17(6).

• Rojhannezhad M. Functional and Bioinformatic Investigation of HER2-Enhancer1 Using CRISPR/Cas9 Technology. *Sarem Journal of Medical Research*. 2023.

Manuscripts Under Review

• Rojhannezhad M, et al. Replicate-Dependent Transient Association of PRIMA-1 with the p53 L1/S3 Pocket: Insights from Four Independent Molecular Dynamics Replicates and a Negative-Control Simulation. *Scientific Reports*. Under review. First and corresponding author.

• Rojhannezhad M, et al. GPCR Signaling in Adipose Tissue and Liver: Implications for Lipid Homeostasis and Disease. *Biomedicine & Pharmacotherapy*. Third author.

Selected Training, Honors & Languages

Clinical Cytogenetics workshop (University of Tehran School of Medicine, 2014) • Genetic Counseling certificate course (ACECR, 2015) • Next-Generation Sequencing workshop (2017) • Non-coding RNA training (2017) • Mammalian stable cell-line development workshop (2017) • Clinical analysis of WES data (2018).

Peer reviewer, *Journal of Cancer and Tumor International*. Ranked 8th nationally in Iran's Ph.D. entrance examination in Medical Genetics (2014) and 13th in Molecular Genetics (2017). Persian: native; English: professional working proficiency.